Project Proposal

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Background:

Application Domain

As university students, we frequently sign up for new bank accounts, credit cards, or digital subscriptions (like Spotify or Amazon Prime) to take advantage of student discounts or sign-up bonuses. However, once the promotional perks expire or a better offer comes along, many of us simply close the accounts. In the business world, this phenomenon is known as "customer churn."

The application domain of this project lies in Financial Data Analytics and Predictive Modeling for Customer Relationship Management. Specifically, it focuses on using demographic, product usage, and financial behavior data to predict whether a customer will leave a bank or stay. In the highly competitive financial sector, acquiring a new customer is significantly more expensive than retaining an existing one. Therefore, building robust machine learning algorithms to identify high-risk customers before they leave is a critical area of research with massive real-world economic value.

Project Goal

In alignment with the CS289A Path 2 guidelines, the overarching goal of this project is to **build**, rigorously **evaluate**, and **compare** a complete machine learning pipeline that classifies bank customer churn using the Kaggle S4E1 dataset.

Rather than relying on a single architecture, our objective is to explore the methodology design space by implementing and contrasting a diverse set of models. Specifically, we will:

- **Data Curation & Engineering**: Perform exploratory data analysis and rigorous feature engineering on the Kaggle dataset to handle categorical variables (e.g., Geography, Gender) and scale continuous features (e.g., Balance, Estimated Salary).
- **Methodological Diversity**: Train multiple classes of models, ranging from classical machine learning algorithms (like Logistic Regression and tree-based models such as XGBoost/Random Forest) to deep learning models that utilize gradient descent via backpropagation (Multi-Layer Perceptrons/Neural Networks).
- **Rigorous Evaluation**: Conduct ablation studies on our neural network architectures to justify our design decisions.

- **Trade-off Analysis**: Contextualize our final model by comparing the predictive performance, using metrics like AUC-ROC and F1-score for imbalanced data, against computational requirements and model interpretability across all the approaches we tested.

Data/Resources:

We plan to use the **Bank Customer Churn** dataset from the Kaggle Playground Series S4E1. The dataset contains **165,034 labeled samples** and **10 input features**, including customer age, gender, geography, credit score, account balance, tenure, number of products, credit card ownership, activity status, and estimated salary. The **target label is Exited**, which indicates whether a customer has left the bank (1) or stayed (0). The dataset also includes three identifier columns (id, CustomerId, and Surname), which we plan to exclude from training.

The dataset is synthetically generated and publicly available on Kaggle. We can download it for free after joining the competition and accepting its rules. We will use the labeled training data and split it into training, validation, and test sets for our experiments.

We plan to use Python, scikit-learn, and PyTorch to preprocess the data, train models, and evaluate their performance. We will run our experiments on Google Colab and our personal laptops. Since the dataset is relatively small and the models we plan to use are not very large, we do not expect to need specialized hardware or paid computing resources.

Timeline:

Individual work:

- **Qizhi:** Lead data exploration and preprocessing, create shared data splits, and implement and tune logistic regression and Linear SVM.
- **Junyu:** Implement and tune random forests, and lead experiments on feature groups and training data size.
- **Lorraine:** Implement and tune an MLP trained through backpropagation, and test different architectures and dropout settings. Explore a ResNet-style MLP if time allows.

Shared work:

- Compare models using accuracy, F1, ROC-AUC, and training time.
- Review experimental results, analyze prediction errors, and choose final model configurations together.
- Read related work and discuss how our findings compare with existing methods.

- Contribute to the proposal, progress report, final paper, and video, with each member documenting their own experiments.
- Meet weekly to review progress, address problems, and update the plan.

Timeline:

Oct 10–16

- Explore the dataset and check missing values, duplicates, and class balance.
- Create shared training, validation, and test splits.
- Complete the preprocessing pipeline.

Oct 17–23

- Train baseline logistic regression, random forest, Linear SVM, and MLP models.
- Compare accuracy, F1, and ROC-AUC using the same data splits.
- Obtain initial results for all four models.

Oct 24–30

- Tune model hyperparameters using validation data.
- Test different MLP architectures and dropout settings.
- Compare prediction performance and training time.

Oct 31–Nov 6

- Summarize preliminary results and each member's contributions.
- Create a shared outline for the final paper.
- Identify remaining experiments.
- **Submit the progress report by Nov 6.**

Nov 7–13

- Test how removing different feature groups affects performance.
- Compare models trained on different amounts of training data.
- Evaluate the effects of MLP architecture and dropout.
- Analyze common prediction errors.
- Complete the main experiments and ablation studies.

Nov 14–20

- Select final model configurations using validation results.
- Evaluate the selected models on the held-out test set using accuracy, F1, and ROC-AUC.

- Prepare tables and figures comparing performance, training time, and ablation results.
- Complete a full paper draft by Nov 20.

Nov 21–27

- Review and revise the paper together.
- Check that the experiments and results are reproducible.
- **Submit the first draft by Nov 24.**
- Begin planning the final video.

Nov 28–Dec 4

- Each member completes their assigned peer review.
- **Submit peer reviews by Dec 4.**
- Prepare the first version of the final video.

Dec 5–11

- Address reviewer feedback and complete any necessary follow-up analysis.
- Polish the final paper, reviewer response, and video.
- Finish all materials by Dec 9 for a final group check.
- **Submit the final report and 2–3 minute video by Dec 11.**